

Response to Request for Information: Development of an Artificial Intelligence (AI) Action Plan

Submitted by: Thomas E. Cheatham, III

Organization: Campus Research Computing Consortium (CaRCC)

Date: 3/14/2025

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Introduction

The Campus Research Computing Consortium (CaRCC) appreciates the opportunity to provide input to the Office of Science and Technology Policy (OSTP) regarding the development of the AI Action Plan as directed by Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence). CaRCC represents professionals who develop, deploy, and support advanced computing and data infrastructure across U.S. academic and research institutions. Our community plays a critical role in enabling AI research, education, and innovation nationwide, and we are committed to sustaining and enhancing America's global AI dominance to promote human flourishing, economic competitiveness, and national security.

Priority Policy Recommendations for the AI Action Plan

1. Establish a National Research Computing and Data Professional Workforce Initiative

Recommendation: Develop and fund a comprehensive national initiative to expand, diversify, and strengthen the research computing and data (RCD) professional workforce.

Rationale: The United States' ability to maintain global AI dominance is directly dependent on having skilled professionals who can build, maintain, and advance the computational infrastructure required for AI development without unnecessary regulatory barriers. Currently, there is a critical shortage of qualified RCD professionals across academia, industry, and government. Without addressing this workforce gap, America's AI capabilities will be severely constrained, hampering both economic competitiveness and national security.

Specific Actions:

- Create dedicated federal funding streams for universities and research institutions to recruit, train, and retain RCD professionals
- Establish formal career pathways and professional development opportunities for RCD professionals, including those specializing in AI infrastructure
- Develop standardized competency models and credentials for RCD professionals that align with industry needs
- Reduce bureaucratic hurdles that prevent the efficient hiring and retention of technical talent
- Establish fellowship programs to attract top talent from diverse backgrounds to RCD careers

2. Create a National Research Computing and Data Infrastructure Program for AI

Recommendation: Establish a coordinated national program to build, sustain, and enhance advanced computing infrastructure specifically designed to support AI research and development.

Rationale: Leading-edge AI research requires substantial computational resources, specialized hardware (like advanced GPUs), massive datasets, and most importantly expert people support. Many academic institutions lack adequate infrastructure to support cutting-edge AI research, creating significant barriers to innovation. The free market alone cannot efficiently address this infrastructure gap, particularly for fundamental research that underpins later commercial applications. A coordinated national approach would remove these barriers while preserving the entrepreneurial spirit that drives American AI leadership.

Specific Actions:

- Create a national network of AI-ready computing facilities with tiered access for researchers across the country
- Establish sustainable funding models for the acquisition, operation, and timely refresh of specialized AI computing hardware
- Develop shared data repositories and pipelines optimized for AI training and testing
- Support energy-efficient computing solutions for AI workloads
- Develop a cadre of RCD professionals to support these goals

3. Integrate Research Computing and Data Professionals into AI Education and Workforce Development

Recommendation: Formally incorporate research computing and data professionals in AI education and training programs at all levels.

Rationale: Teaching AI concepts without practical, hands-on experience with real AI systems limits the effectiveness of education programs. Research computing and data professionals possess the practical knowledge needed to bridge theoretical concepts with real-world implementation.

Specific Actions:

- Fund embedded research computing and data professionals within academic departments offering AI education
- Develop course materials and curricula that integrate practical AI infrastructure skills
- Create apprenticeship and internship programs that pair students with research computing and data professionals
- Establish programs to transition skilled professionals from other sectors into research computing and data roles

4. Remove Barriers to Public-Private Collaboration in AI Infrastructure

Recommendation: Eliminate regulatory obstacles and establish streamlined policy frameworks that facilitate collaboration between academic research computing and data organizations and private industry.

Rationale: Effective AI innovation requires seamless collaboration between academic researchers and industry. Current administrative, legal, and regulatory barriers—many created by previous policies now revoked by Executive Order 14179—have prevented the efficient sharing of resources, knowledge, and talent. Removing these barriers will unleash American innovation and entrepreneurship in the AI space.

Specific Actions:

- Create streamlined agreements for public-private partnerships in AI infrastructure development
- Establish tax incentives for industry donations of computing hardware, software, and expertise to academic institutions
- Develop fellowship programs that enable professionals to move between industry and academic research computing and data roles
- Create regulatory frameworks that facilitate responsible data sharing between sectors

5. Support Research Computing and Data as a Critical Component of National Security

Recommendation: Recognize and support research computing and data infrastructure and professionals as essential to national security and American AI dominance.

Rationale: As AI becomes increasingly important to national security, the professionals who develop and maintain AI infrastructure must be viewed as strategic assets. Currently, these

roles often lack the recognition, support, and protection afforded to other critical infrastructure sectors. Strengthening this workforce directly supports the national security objectives outlined in Executive Order 14179.

Specific Actions:

- Include research computing and data infrastructure in critical infrastructure protection frameworks
- Develop specialized training and career paths for research computing and data professionals working on national security applications
- Create rapid response capabilities for research computing and data professionals to address emerging AI security challenges
- Establish secure environments for sensitive AI research that maintain academic freedom while protecting national interests

Conclusion

The United States' global dominance in artificial intelligence depends not only on algorithmic breakthroughs and research innovations but also on the robust infrastructure and skilled professionals who make that research possible. By investing in and empowering the research computing and data professional workforce while removing unnecessary regulatory barriers, the AI Action Plan can establish a foundation for sustainable American leadership in artificial intelligence that promotes human flourishing, economic competitiveness, and national security.

The recommendations presented here align directly with Executive Order 14179's goal of removing barriers to American leadership in AI. By addressing critical workforce and infrastructure needs while maintaining a light regulatory touch, we can ensure America maintains its competitive edge in this transformative technology.

CaRCC stands ready to collaborate with OSTP, the Special Advisor for AI and Crypto, and other stakeholders to implement these recommendations and contribute to the development of a comprehensive AI Action Plan that ensures America's continued global dominance in artificial intelligence.

Contact Information:

Thomas E. Cheatham, III
Professor / Director / CaRCC Chair
Campus Research Computing Consortium (CaRCC)

[REDACTED] or [REDACTED]
[REDACTED]